

Question number	Answer	Mark
4(a)(i)	The only correct answer is D <i>A is incorrect because the magnification is 2.67×10^1</i> <i>B is incorrect because the magnification is 2.67×10^1</i> <i>C is incorrect because the magnification is 2.67×10^1</i> <i>Magnification = $48 \text{ mm} \div 1.8 \text{ mm} = 26.66667 = 2.67 \times 10^1$</i>	(1)

Question number	Answer	Mark
4(a)(ii)	The only correct answer is B <i>A is incorrect because the surface area is 1.8π</i> <i>C is incorrect because the surface area is 1.8π</i> <i>D is incorrect because the surface area is 1.8π</i> <i>$SA = 2\pi \times 0.5 \times 1.8 = 1.8\pi$</i>	(1)

Question number	Answer	Additional guidance	Mark
4(a)(iii)	• volume calculated (1) • 1 : 0.3 / 1 : 0.28 (1)	$\pi \times 0.6 \times 0.6 \times 1.8 = 0.648\pi$ ACCEPT 1.944 / 2.03575203952 NB if clearly used a non-calculator value for π check the rounding before awarding ACCEPT 4 : 1 / 3.5 : 1 / 3.55 : 1 (3.5493827160) DO NOT ACCEPT 3.6 : 1 ACCEPT CEs for $V = 2.592\pi$ 1 : 1.1 / 1 : 1.13 0.9 : 1 / 0.89 : 1 Correct bald answer = 2 marks Correct bald CE = 1 mark	(2)

Question number	Answer	Additional guidance	Mark
4(b)	An explanation that makes reference to three of the following points: - oxygen {dissolved / present} in the water (1) - large surface area (to volume ratio) for fast {gas exchange / diffusion (of gases)} (1) - two {layers (of cells) / cells thick} so {short diffusion distance (for gases) / fast gas exchange} (1) {tentacles / body} move the water around to maintain (oxygen) concentration gradient (1)	IGNORE efficient and easy throughout NB penalise refs to {lungs / alveoli / blood / vessels / etc} within an awardable mp 2, 3 or 4 once ACCEPT more diffusion IGNORE hollow body / tentacles ACCEPT thin layer of cells so ACCEPT other terms which clearly refer to tentacles e.g. tails, legs, fingers	(3)